

$$S_0 = \mathbb{R} \oplus$$

$$\begin{aligned} SB &= \sqrt{S_0^2 + B_0^2 - 2S_0 \cdot B_0 \cdot \cos \theta} \\ &= \left[R_\oplus^2 + (R_\oplus + H)^2 - 2R_\oplus(R_\oplus + H)\cos \theta \right]^{\frac{1}{2}} \\ &= \left\{ (1 - \cos \theta) [2R_\oplus(R_\oplus + H)] + H^2 \right\}^{\frac{1}{2}} \end{aligned} \quad \left. \begin{array}{l} \text{Cosine} \\ \text{law} \\ \text{on } \triangle SBO \end{array} \right\}$$

$$AB^2 = SB^2 + AS^2 + SB \cdot AS \cdot \cos(90^\circ - \varphi)$$

$$\sin(\varphi) = \frac{AB^2 - SB^2 - AS^2}{SB \cdot AS}$$

$$= \frac{(R_\oplus + H)^2 \cdot 2(1 - \cos\theta) - (1 - \cos\theta)[2R_\oplus(R_\oplus + H)] + \cancel{H^2} - \cancel{H^2}}{SB \cdot AS}$$

$$= \frac{2(R_\oplus + H) \cdot (1 - \cos\theta) [\cancel{R_\oplus + H} - \cancel{R_\oplus}]}{SB \cdot \cancel{H}}$$

$$\Rightarrow \sin \varphi = \frac{2(R_\oplus + H) \cdot (1 - \cos\theta)}{\sqrt{(1 - \cos\theta)[2R_\oplus(R_\oplus + H)] + H^2}}$$

$$\approx \sqrt{\frac{2(R_\oplus + H) \cdot (1 - \cos\theta)}{R_\oplus}}$$

$H \ll R$

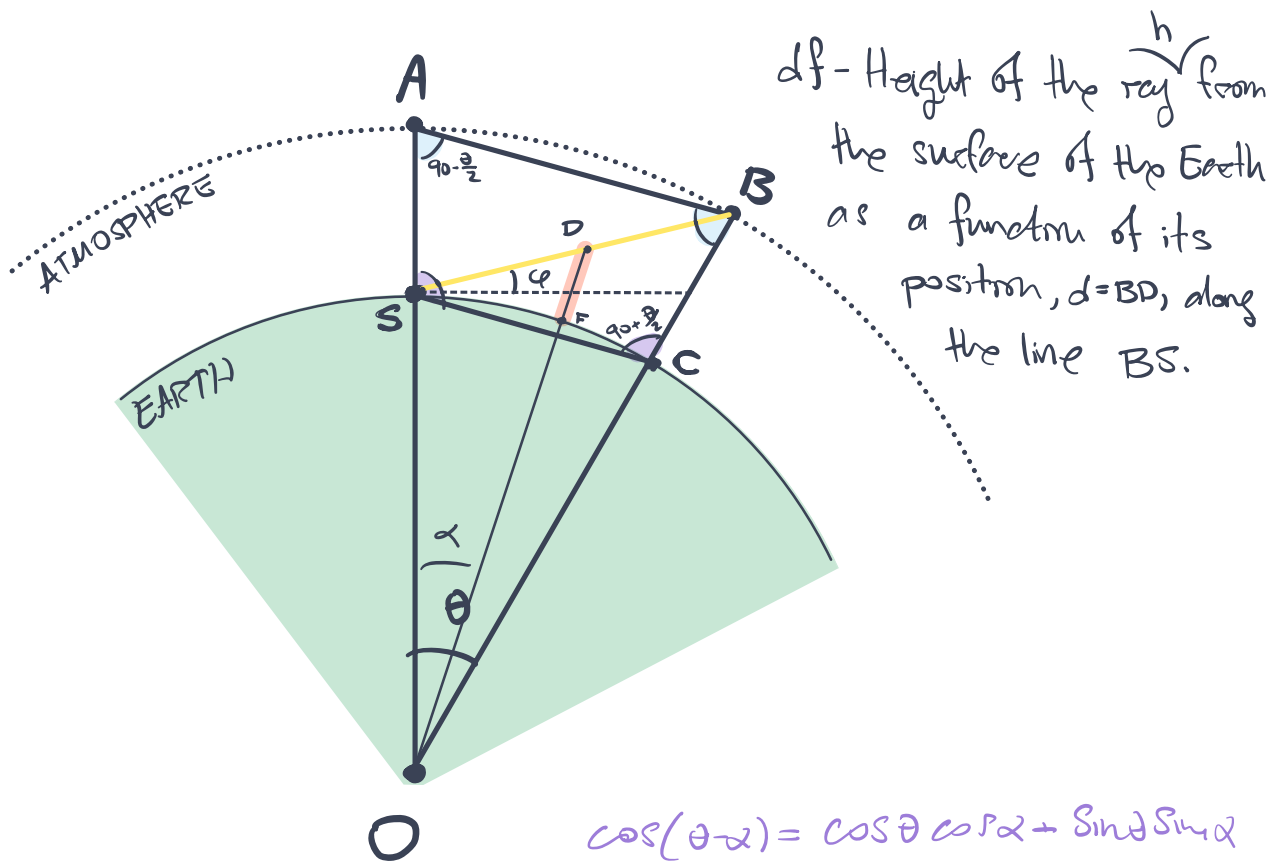


$$(1 - \cos \theta) [2R_{\oplus}(R_{\oplus} + H)] + H^2$$

$$= (1 - \cos \theta) [2R_{\oplus}^2 + 2R_{\oplus}H] + H^2$$

$$\Rightarrow R_{\oplus}^2 \left\{ (1 - \cos \theta) \left[2 + \frac{H}{R_{\oplus}} \right] + \left(\frac{H}{R_{\oplus}} \right)^2 \right\} \approx (1 - \cos \theta) [2R_{\oplus}(R_{\oplus} + H)]$$

$$\begin{aligned} H &\propto 10^1 \\ R_{\oplus} &\propto 10^3 \Rightarrow \left(\frac{H}{R_{\oplus}} \right)^2 \propto 10^{-6} \end{aligned}$$



$$\triangle OBD : DB^2 = OB^2 + OD^2 - OB \cdot OD \cdot \cos(\theta - \alpha)$$

$$d^2 = (R_0 + H)^2 + (R_0 + h)^2 - (R_0 + H)(R_0 + h) \cdot \cos(\theta - \alpha)$$

$$\triangle OSD : SD^2 = OS^2 + OD^2 - OS \cdot OD \cdot \cos(\alpha)$$

$$(1) \quad \cos(\alpha) = \frac{SD^2 - OS^2 - OD^2}{OS \cdot OD}$$

(2)

$$DB^2 = OB^2 + OD^2 - OB \cdot OD \cdot [\cos \theta \cos \alpha + \sin \theta \sin \alpha]$$